

Semi-Autonomous Wildfire Monitoring and Control System Framework

Gene Foxwell

Woolf University / Udacity School of Artificial Intelligence

Industry Integrated AI-Synthesis Project

Overview and Industry Context

In recent decades, climate change has increased wildfire vulnerability in forests across many regions, including Canada (Wang 2025). As temperatures continue to rise, this threat is unlikely to decline soon. Wildfire response requires substantial resources and creates serious risks to human life (Belval 2025). This paper proposes a framework for a semi-autonomous wildfire monitoring and control system. It draws on techniques from previous projects, explaining how they are integrated and why they were selected. The paper then presents the system design, examines ethical considerations, reviews framework testing results, and discusses the project's relevance for organizations seeking AI professionals.

Integration Rationale

Three primary AI techniques were used in the design of this project:

1. **Semantic Segmentation:** We wanted a fast way to provide the system with predictions of where the fire would go next. Assuming we had either a high-altitude aerial view, or at a periodic satellite view, we built a “risk” map of the area would be a good approach. This is a simple binary classification problem for each cell in our simulation. We reused the UNet (Ronneberger 2015) built for Project 4 here.
2. **Agentic AI:** A Semi-Autonomous command center was needed to provide commands to the UAVs. We chose to build this using Agentic AI with GraphRAG (Han 2025). A similar approach was used in Project 6.

3. **Data Analysis:** We collect data on the systems performance and use the Man-Whitney U Test (Mann 1947) to determine if the system had a statistically significant effect on the fire simulation. We used this test to compare asteroid populations in Project 2.

System Design and Technical Decisions

A visual overview of the system design can be found in Figure 5.

Environmental Module

This module was responsible for simulating the environment as well as generating “risk predictions” based on the environment.

Our environment was randomly generated using Perlin Noise, adapted from (Kamble 2025). We assigned environmental features: grasslands, trees, water, housing, and urban areas to numerical ranges. Figure 1 shows an example of the terrain this generates.

A similar approach was used to generate a static model for wind. To keep the simulation tractable, we assumed wind to be static once generated.

We simulated the Wildfire using a Stochastic Cellular Automata adapted from (Ghosh 2024) and (Ramadan 2024). We begin each simulation by igniting a single “cell” and letting the fire play out. Equation 1 determines the probability of fire spreading from one cell to another. An example can be seen in Figure 2.

Equation 1:

$$p_{ignite} = [1 + \exp(-k * burningtrees - windalignment - burnthreshold)]^{-1}$$

VARIABLE	INTERPRETATION
K	A scaling parameter.
BURNINGTREES	The number of burning cells adjacent to this cell.
WINDALIGNMENT	The dot product of the average wind direction and average fire direction about this cell.
BURNTHRESHOLD	A constant determining the minimum probability we need to “ignite” a cell.

Each burnable cell has a finite amount of fuel which decremented at the end of every step. When a cell runs out of fuel, it is converted to ash.

Given that humans sometimes head out into the forest for various reasons, we added human agents to the simulation. These human agents are stochastic and are considered “casualties” when they encounter fire.

Risk Predictor (Sub-Module of Environment Module)

We trained a modified version of a UNet to perform risk prediction. We altered the bottleneck to use multiple 3x3 convolutional layers instead of a single 1x1 convolution. Figure 3 shows a schematic diagram of our UNet.

We created a synthetic dataset using simulation output. We recorded a random starting point in the simulation, ran it for 20 steps, then recorded the output. Areas in the final image that

were still burning or destroyed (but not destroyed in the original image) were marked “Risky” in the mask. An example of the results after training can be seen in Figure 4.

Model Training Parameters:

PARAMETER	VALUE
EPOCHS	10
LEARNING RATE	0.001
OPTIMIZER	AdamW
LEARNING RATE SCHEDULER	Mode=minimization, factor=0.5, patience=3

We used a hybrid loss function, as shown in Equation 2

Equation 2:

$$loss = 0.5 * BCELossWithLogitsLoss + 0.5 * DiceLoss$$

The BCEWithLogitsLoss guides the system towards handling binary classification on a per-pixel basis, while the DiceLoss helps ensure that the overall shape of the output is preserved. This helps mitigate class imbalance. Early stopping was used to avoid overfitting. Loss curves can be seen in Figure 7.

CLASS & METRIC	SCORE
SAFE – MIOU	0.9422
SAFE – DICE	0.9698

RISKY – MIOU	0.6205
RISKY - DICE	0.7564

UAV Module

We define three UAV types: Recon (a direct analogue to the UAV in Merlin 2009), Extinguishers equipped to put out fires, and SAR UAVs that can pick up humans from unsafe areas.

Subsumption architecture (see Figure 8) was used to model UAV behavior. Connell 1990 provides an in-depth example of this approach.

1. Boundary Override moves UAV away from boundaries.
2. Return Home moves UAV to home base when low on fuel.
3. Head to waypoint moves UAV to waypoint.
4. Heuristic Module is activated if there is no waypoint.

UAVs keep a record of all observations / positions / waypoints for each step. Data is sent to the Command Module after an appropriate number of steps have passed. UAVs accept commands from the Command Module.

Command Module

We used Agentic Workflow as the basis for our Command Module. Figure 6 provides a schematic diagram of the workflow.

Each Agent other than the Command Centre Agent keeps a memory of previous interactions which are inserted as part the prompt for each turn. Memory is implemented as a deque. To avoid exceeding the input token limit on our model, we truncate the input. Historical data is fed to the agent in reverse chronological order.

Long term goals (LTG) are tracked using a Knowledge Graph, adapted from (Yang 2025). We have three LTG's: Search and Rescue, Housing Protection, and Fire Fighting. Our KG ontology is in Figure 6.

MGraphDB¹ was used for GraphDB. We created tools for the Agents to add/remove/query entities and relationships from the graph.

AGENT	PURPOSE	KNOWLEDGE GRAPH ACCESS
Command Centre Agent	Orchestration. The command center agent takes in the inputs from the simulation and outputs the final response from the Workflow.	None
Strategy Agent	Generate Subgoals for LTGs	Read/Write
Dispatch Agent	Translate the activities and priority list generated by the Strategy Agent UAV Commands	Read

¹ [mgraph-db · PyPI](#)

Assistant Agent	Responds to user queries and updates the LTG in the Knowledge Graph as needed.	Read/Write
------------------------	--	------------

Integration worked as follows:

1. On the first step and every 40-simulation step afterwards², the simulation calls the Command Center Agent. The simulation provides the recent history of each UAV, a screenshot of the current area, and output from the UNet model.
2. On the first invocation of the Command Center Agent, and every 5 iterations afterwards, the Strategy Agent is invoked to update LTGs.
3. The most recent priorities and inputs are sent to the Dispatcher Agent. The dispatch agent produces a new waypoint for each UAV under its control. A reason for each command provided.
4. These commands are passed on to the UAVs.

The Operator can interrupt the simulation while it is running to query the current state. These queries are routed to the Assistant Agent, which answers questions and makes updates to the Knowledge Graph as needed.

The simulation ends either after 750 steps, or when all fire has been extinguished.

² Optimizes runtime and token usage.

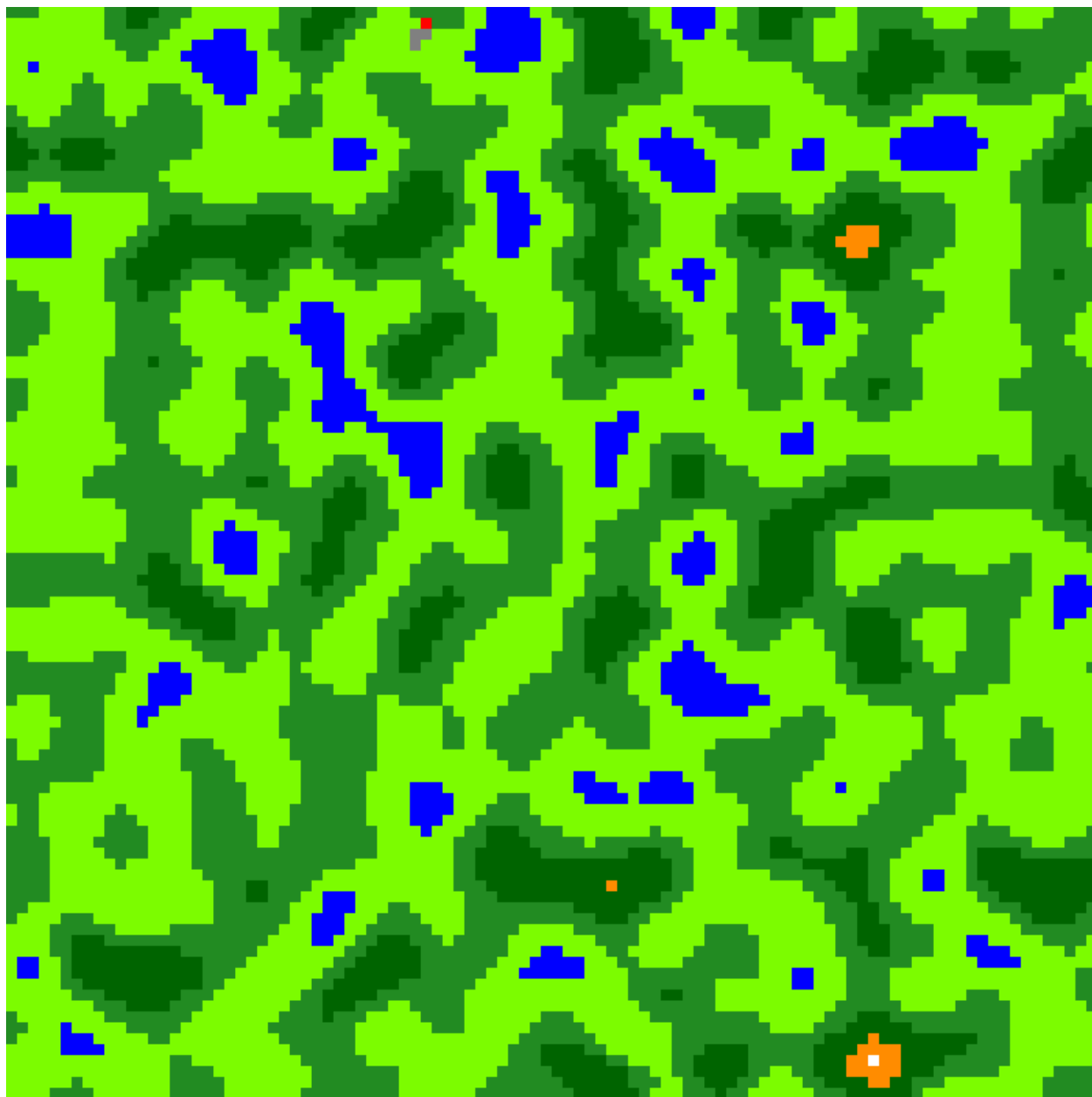


Figure 1 Perlin Noise Generated Terrain

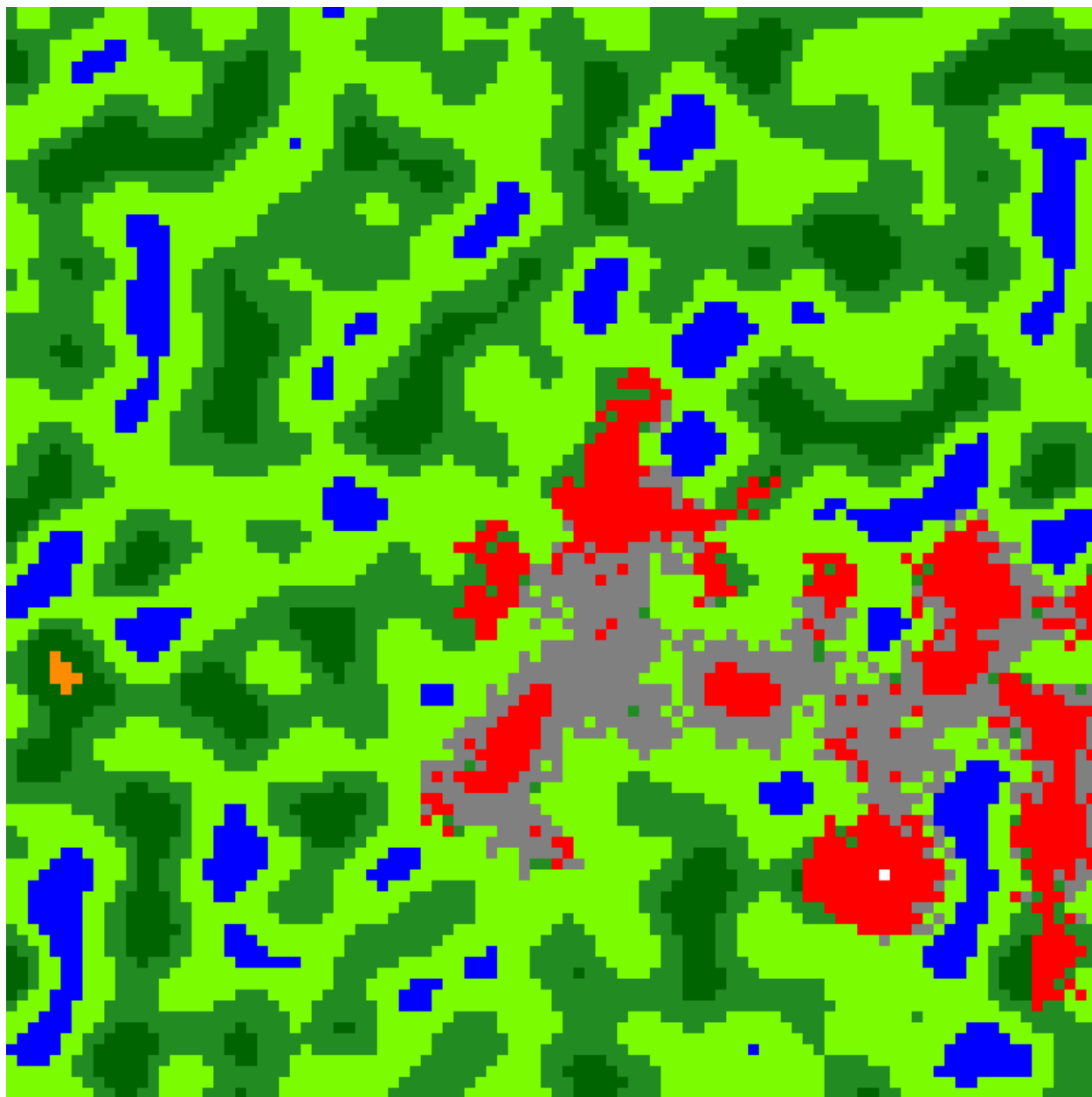
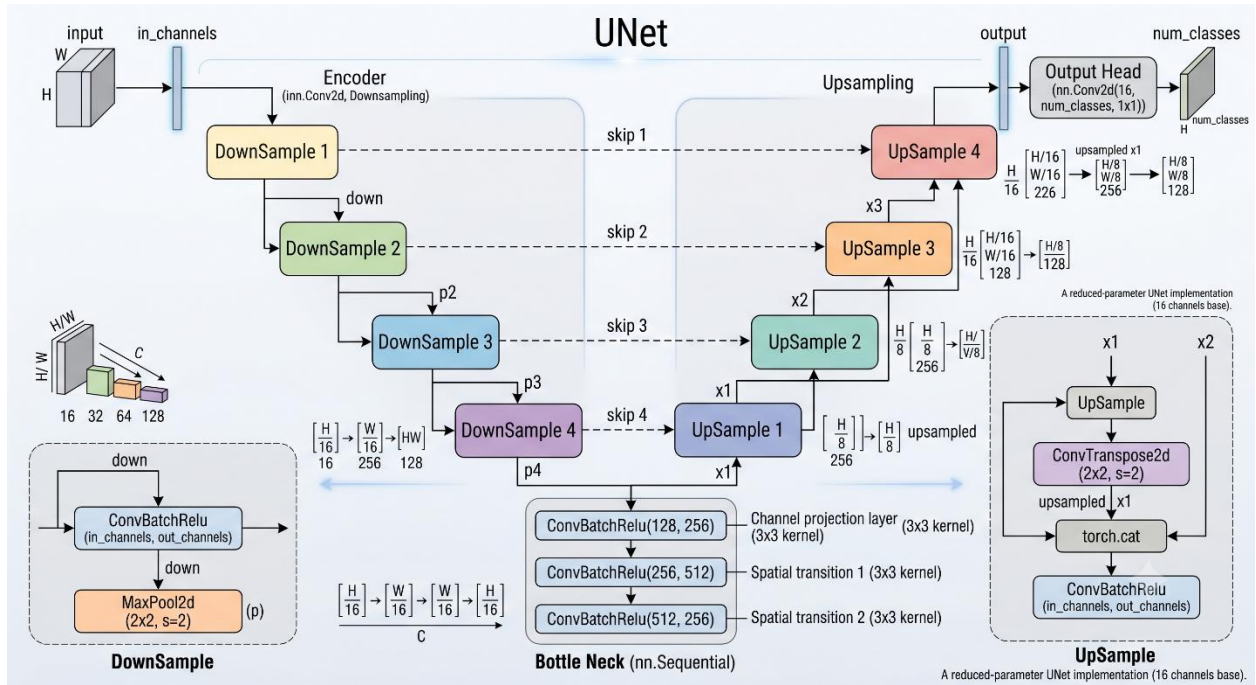


Figure 2 Wildfire CA in action



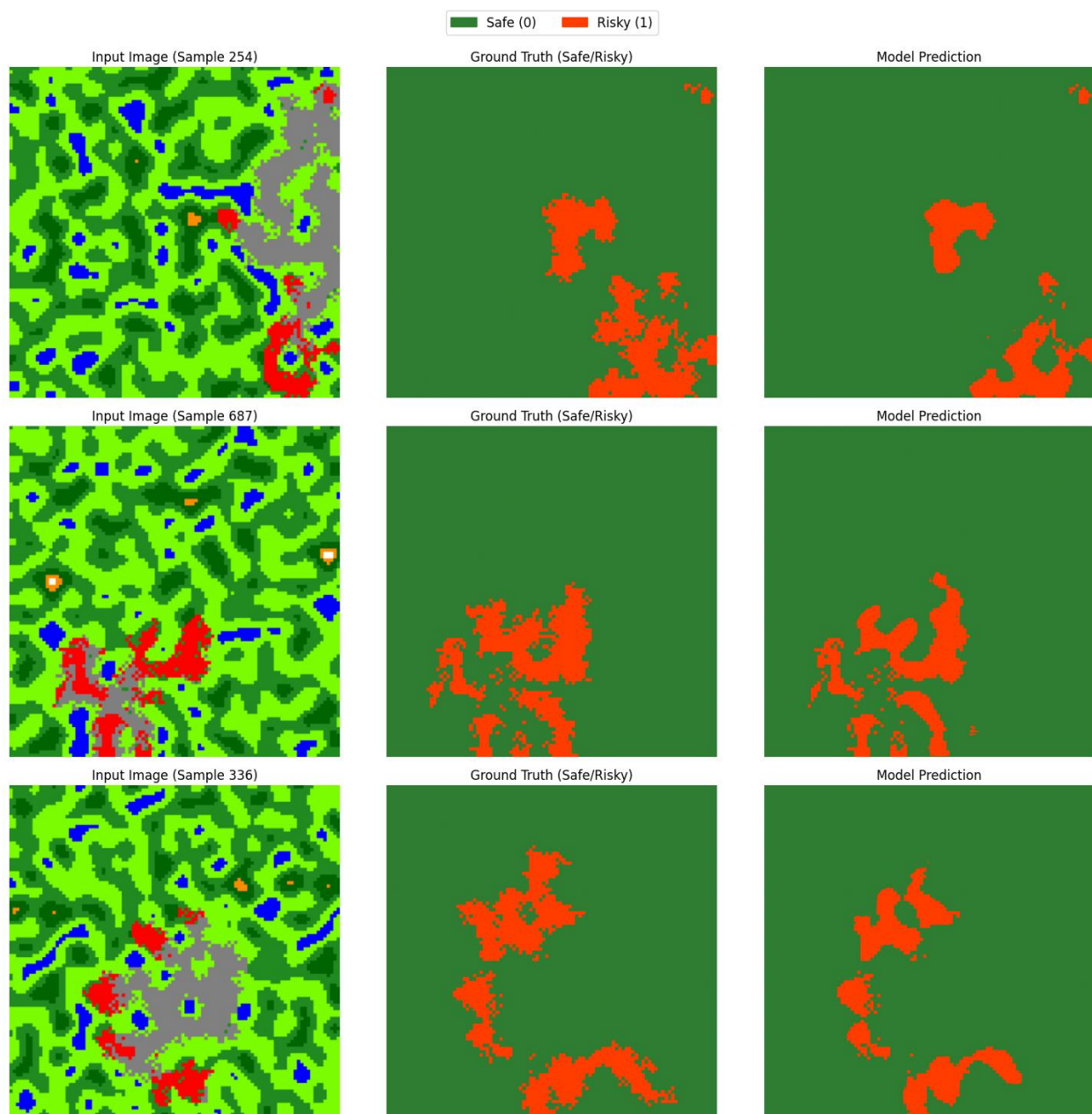


Figure 4 Fire Risk Prediction Outputs

Semi-Autonomous Wildfire Monitoring & Control System Framework Gene Foxwel 2026

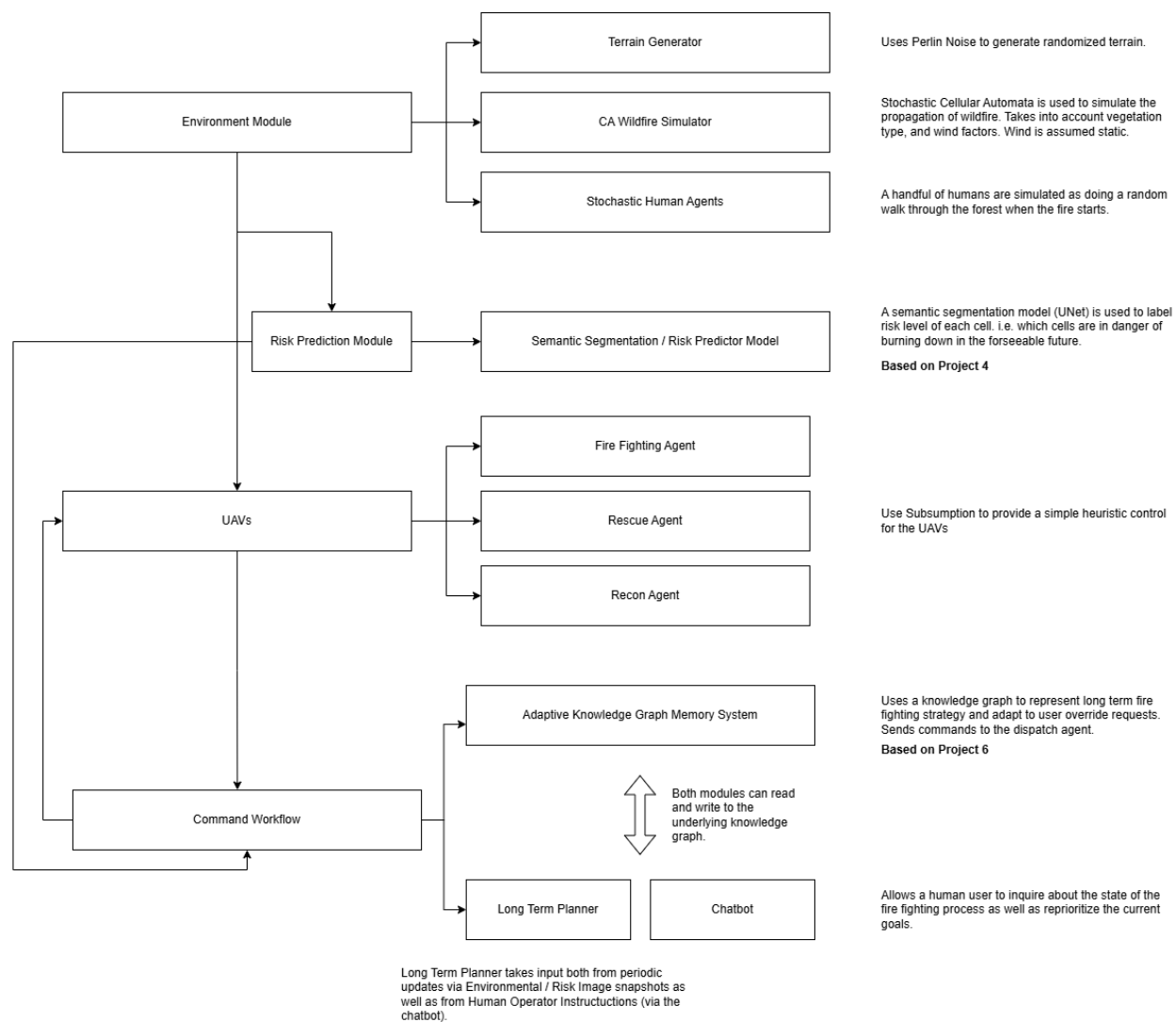
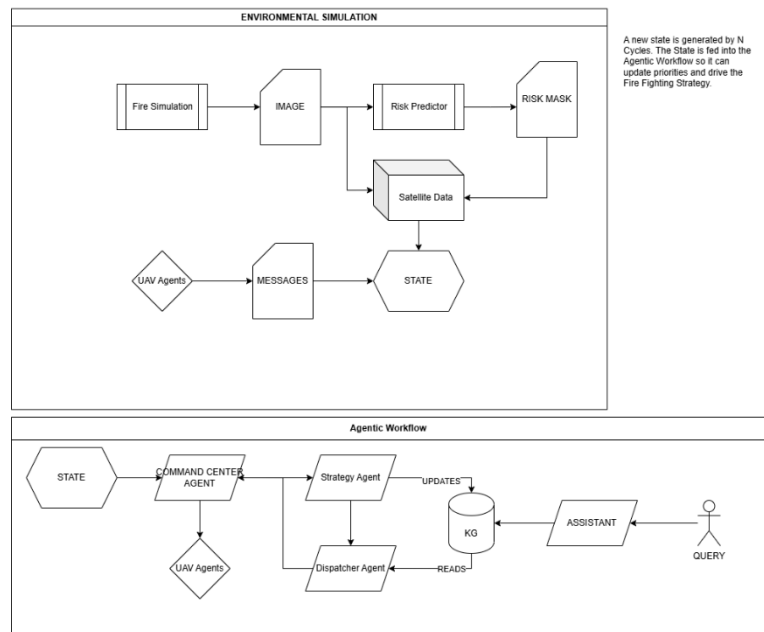


Figure 5 System Overview

Wild Fire Command and Control Workflow



ONTOLOGY

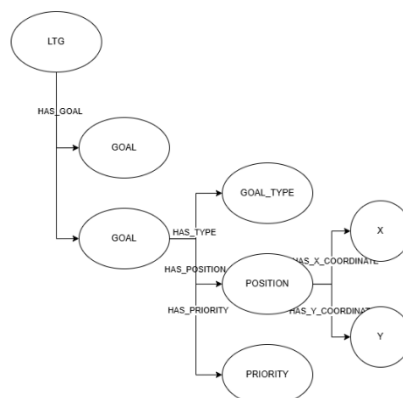


Figure 6 Agentic Workflow

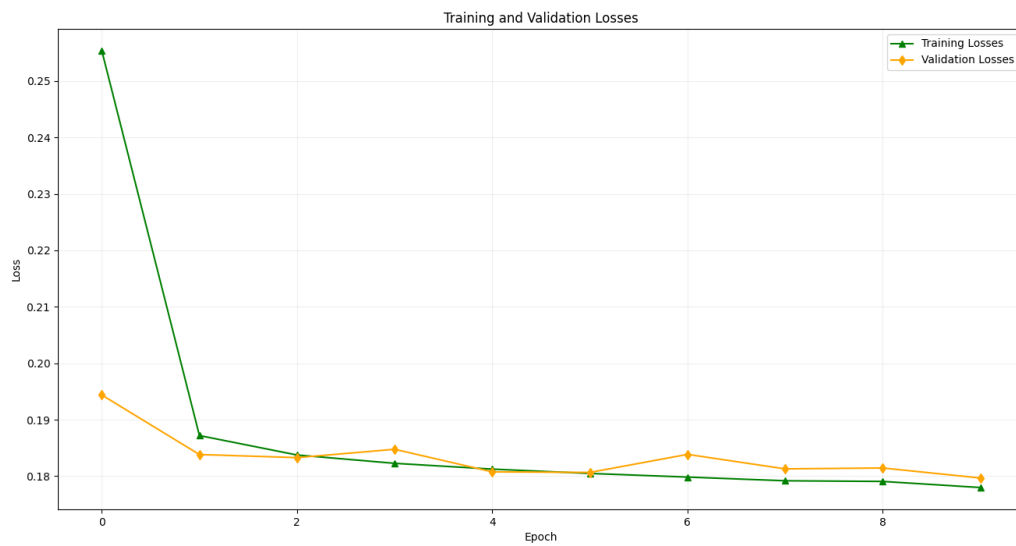


Figure 7 UNet Training and Validation Loss

UAV Behavior Diagram

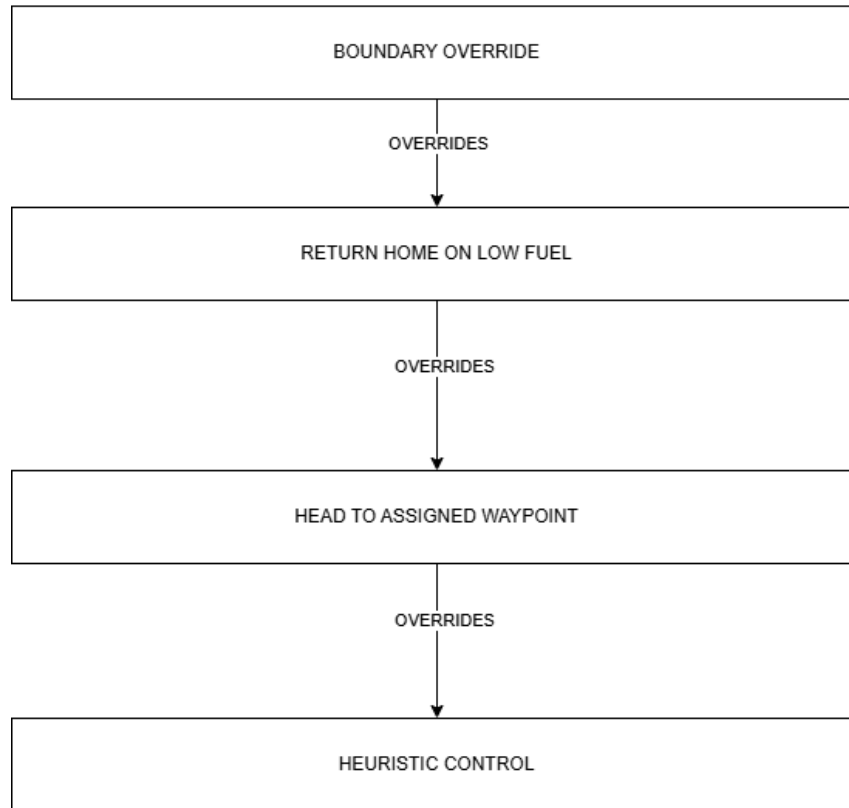


Figure 8 UAV Behavior

Ethical Considerations and Responsible AI

While fighting wildfires is a dangerous occupation, it still provides employment. Replacing these workers could leave them without a comparable source of income. This is not an intended outcome of this framework. This work and similar work should augment human abilities.

We do not recommend extending this system to be fully autonomous. Agents can sometimes fixate on unexpected things. An Agent that continues to search for a dead human while ignoring others (as happened during testing) is not a desirable situation.

Assumptions were made during testing that may not be realistic.

Examples:

1. Static wind field. Wind is rarely static. We assumed reliable aerial views of the fire.
2. Firefighting and SAR drones could be built reliably. Furthermore, especially for SAR, a human presence is preferable to deal with first-aid or psychological distress that might occur during an emergency.

Who is responsible for loss of life / property damage that may occur due to a system like this? A malfunction or a choice could lead to loss of life or property. We left in the human-in-the-loop option to override the system, but people will still defer to the computer, especially when stressed. Protocols will need to be in place to review the reasoning presented by the system to ensure they meet the expected standards.

Our setup used a remote model – Anthropic Claude Sonnet 4.5. This was sufficient for our purposes. Infrastructure may not be reliable in a realistic scenario. A local model should be used (for example Llama 4 Scout, Minstral Small³) so the system is independent of local infrastructure.

Evaluation and Reflection

We measured the effectiveness of this system by performing the following experiment:

1. Run the simulation 30 times without the Agentic Workflow and UAV active.
2. Run the simulation 30 times with Agentic Workflow and UAV.
3. Collected Mean and STD of Burnt Area and Casualties over both datasets.
4. Performed the Man-Whitney U Test (reference) to see if there was a statistically significant difference.

³ [The Best Open Source and Open-Weight LLM Models to Run Locally in 2026](#)

Figure 9 shows the burned area and casualties for each of our two datasets. The table below provides the results of Man-Whitney U test.

FEATURE	BASE MEDIAN	AGENTIC MEDIAN	P-VALUE
BURNED AREA	22.9%	13.45%	0.0023
CASUALTIES	5	3	0.012

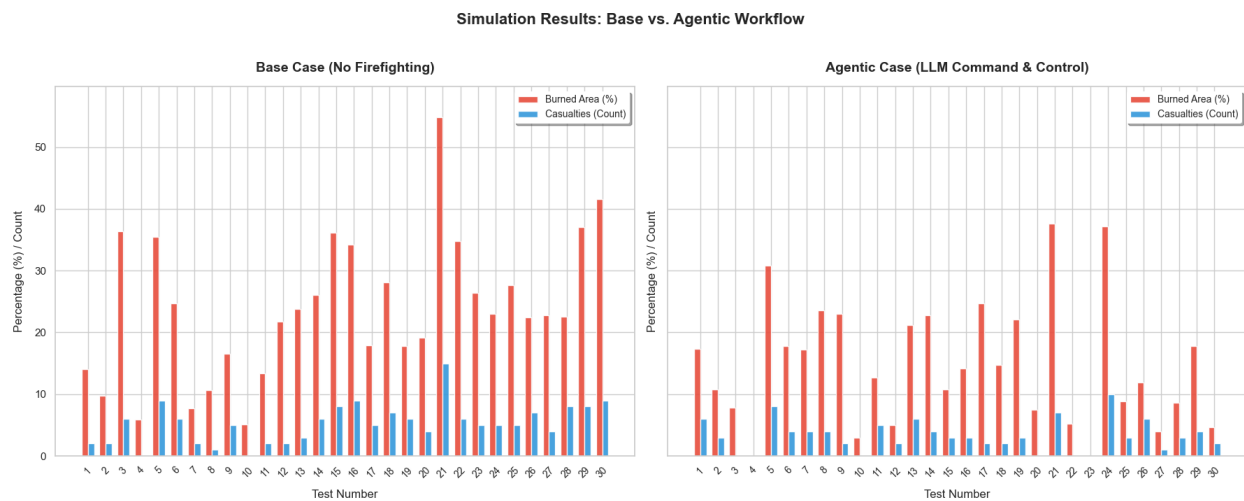


Figure 9 Simulation Results

The table shows $p\text{-values} < 0.05$ for both features we compared. This shows that our system reduced the burned area as well as the number of casualties by a statistically significant margin.

Professional Relevance

This project demonstrates several techniques in Artificial Intelligence:

1. Semantic Segmentation as a predictive technique – which could have potential applications in Autonomous Driving (Feng 2021) and Robotics (Hurtado 2022) (as well as the original application of UNet to medical imagery).
2. Agentic Workflows, which is finding applications in Finance (Yang 2024), and Entertainment (Ammanabrolu 2020).
3. Statistical Analysis to show our approach can have a positive effect on the outcome.

We combined these to model Wildfire Monitoring Control, demonstrating the ability to synthesize multiple techniques into a single system.

References

Ammanabrolu, P., Cheung, W., Tu, D., Broniec, W., & Riedl, M. O. (2020). Bringing stories alive: Generating interactive fiction worlds. *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*.

<https://arxiv.org/pdf/2001.10161>

Belval, E. J., Pietruszka, B. M., & Viktora, A. (2025). The price of doing business: Severe injuries in wildland firefighters in the United States by activity performed and hazard encountered. *International Journal of Wildland Fire*.

<https://research.fs.usda.gov/treesearch/69384>

Centers for Disease Control and Prevention. (2014). Aviation-related wildland firefighter fatalities — United States, 2000–2013. *Morbidity and Mortality Weekly Report*.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4584834/>

Chen, L., Lu, K., Rajeswaran, A., Lee, K., Grover, A., Laskin, M., Abbeel, P., Srinivas, A., & Mordatch, I. (2021). *Decision transformer: Reinforcement learning via sequence modeling*. arXiv. <https://doi.org/10.48550/arXiv.2106.01345>

Connell, J. H. (1990). *Minimalist mobile robotics: A colony-style architecture for an artificial creature*. Academic Press.

Feng, D., Haase-Schutz, C., Rosenbaum, L., Hertlein, H., Glaser, C., Timm, Fabian., Wiesbeck, W., & Dietmayer, K. (2021). Deep Multi-Modal Object Detection and Semantic Segmentation for Autonomous Driving: Datasets, Methods, and Challenges. *IEEE Transactions on Intelligent Transportation Systems*, 22(3), 1341-1360.
<https://doi.org/10.1109/tits.2020.2972974>

Ghosh, R., Adhikary, J., & Chemlal, R. (2024). *Fire spread modeling using probabilistic cellular automata*. arXiv. <https://doi.org/10.48550/arXiv.2403.08817>

Han, H., Ma, L., Shomer, H., Wang, Y., Lei, Y., Guo, K., Hua, Z., Long, B., Liu, H., Aggarwal, C. C., & Tang, J. (2025). *RAG vs. GraphRAG: A systematic evaluation and key insights*. arXiv. <https://doi.org/10.48550/arXiv.2502.11371>

Hurtado, J. V., & Valada, A. (2022). Semantic scene segmentation for robotics. *Deep Learning for Robot Perception and Cognition*, 279-311. Elsevier.
<https://doi.org/10.1016/b978-0-32-385787-1.00017-8>

Julian, K. D., & Kochenderfer, M. J. (2018). *Distributed wildfire surveillance with autonomous aircraft using deep reinforcement learning*. arXiv.
<https://doi.org/10.48550/arXiv.1810.04244>

Kamble, A. D., & Ghule, S. D. (2025). Procedural terrain generation using Perlin noise. *International Journal of Innovative Research in Technology*, 12(1), 256–259.

https://ijirt.org/publishedpaper/IJIRT180167_PAPER.pdf

Mann, H. B., & Whitney, D. R. (1947). On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics*, 18(1), 50–60. <https://doi.org/10.1214/aoms/1177730491>

Merlin, P. W. (2009). *Ikhana Unmanned Aircraft System: Western States fire missions* (NASA SP-2009-4544; Monographs in Aerospace History, No. 44). National Aeronautics and Space Administration. <https://www.nasa.gov/wp-content/uploads/2023/03/sp-4544.pdf>

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., Petersen, S., Beattie, C., Sadik, A., Antonoglou, I., King, H., Kumaran, D., Wierstra, D., Legg, S., & Hassabis, D. (2015). Human-level control through deep reinforcement learning. *Nature*, 518(7540), 529–533. <https://doi.org/10.1038/nature14236>

Ramadan, A. (2024). *Wildfire autonomous response and prediction using cellular automata (WARP-CA)*. arXiv. <https://doi.org/10.48550/arXiv.2407.02613>

Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. *arXiv*. <https://arxiv.org/abs/1505.04597>

Wang, W., Wang, X., Flannigan, M. D., et al. (2025). Canadian forests are more conducive to high-severity fires in recent decades. *Science*, 387, 91–97.
<https://doi.org/10.1126/science.adol006>

Watkins, C.J.C.H., Dayan, P. *Q*-learning. *Mach Learn* 8, 279–292 (1992).
<https://doi.org/10.1007/BF00992698>

Yang, H., Chen, J., Siew, M., Lorigo-Botran, T., & Joe-Wong, C. (2025). *LLM-powered decentralized generative agents with adaptive hierarchical knowledge graph for cooperative planning*. arXiv. <https://doi.org/10.48550/arXiv.2502.05453>

Yang, H., Zhang, B., Wang, N., Guo, C., Zhang, X., Lin, L., Wang, J., Zhou, T., Guan, M., Zhang, R., & Wang, C. D. (2024). *FinRobot: An Open-Source AI Agent Platform for Financial Applications using Large Language Models*. arXiv.
<https://doi.org/10.48550/arXiv.2405.14767>